Rainul Hakim

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Education

Davidson College Davidson, NC

Bachelor of Science, Computer Science | GPA: 3.76 *Expected May 2029*

- Relevant Coursework: Data Structures, Database Programming, Linear Algebra, Calculus I & Modeling, Calculus II

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, R, C
ML / Scientific Computing: PyTorch, PyTorch Geometric, TensorFlow.js, MediaPipe, OpenCV, NumPy, Pandas, SciPy, Matplotlib, scikit-learn
Web: React, Next.js, Node.js, Supabase, PostgreSQL, Vercel Serverless Functions, HTML, CSS
Tools: Git, GitHub, VS Code, AWS, Linux, Mapbox, Quarto, HPC clusters

Experience

Machine Learning Research Assistant May 2026 – Aug 2026

Facility for Rare Isotope Beams (FRIB) *East Lansing, MI & Davidson, NC*

- Reached **88.82%** accuracy / **88.60%** weighted F1 on 3-class track classification by pretraining a DGCNN encoder with SimCLR-style contrastive learning on **60,795** unlabeled AT-TPC events, then fine-tuning via linear probing
- Outperformed generic augmentation baselines by designing a physically-motivated rotation scheme, validated through systematic ablation studies
- Trained models with PyTorch and PyTorch Geometric on an HPC cluster; documented code for a cross-institutional, version-controlled research codebase

Data Consultant Aug 2026 – Present

DataCATS (Consulting, Analytics, Tutoring Services), Davidson College

- Led **85**+ consulting sessions and projects for students and faculty on data analysis, statistical modeling, and experimental design
- Debug code and data pipelines and guide regression analysis, hypothesis testing, and visualization

Student Consultant Feb 2026 – May 2026

The Jay Hurt Hub for Innovation and Entrepreneurship

- Built the Next.js/React front end and TypeScript API routes for an AI study planner, delivering a working prototype that drew follow-up interest from Davidson faculty and other consultant teams
- Cut malformed AI output reaching users by implementing Zod-validated response schemas and wellbeing guardrails, verified through iterative QA testing

Projects

Campus Event Discovery Platform | ***Python, PostgreSQL, AWS Lambda, Amazon RDS, API Gateway, Mapbox***

- Built event discovery platform ingesting Davidson event data into a relational **PostgreSQL** database
- Automated AWS pipeline with **EventBridge** and **Lambda** to extract, clean, and update event and location data
- Developed **React/Mapbox** interface to filter events by location, date, and category with campus navigation

Left No Crumbs! – 1st Place ($1,000), IdeaSprint | ***Next.js, Supabase, PostgreSQL, Web Push***

- Co-founded and built a real-time food-rescue platform alerting **600**+ students across **55** organizations to surplus catered meals
- Designed a **PostgreSQL** schema with row-level security, campus-only auth, dietary filtering, and rate-limited push notifications; partnered with Davidson Dining and T&I on Davidson One integration

AI Meal Analysis & Recommendation Platform | ***Python, Gemini API, React, Vite, JavaScript, Vercel***

- Built AI meal analysis pipeline generating macro breakdowns, health flags, and personalized alternatives
- Integrated **Gemini API** behind serverless functions and implemented meal comparison and weekly scoring systems
- Deployed React application used by **100**+ Davidson students to evaluate and compare campus meals

In-Browser Hand Gesture Recognition (ML) | ***JavaScript, TensorFlow.js, MediaPipe*** Open-source extension

- Built a gesture-triggered effects app running fully in-browser at ∼30 fps with no backend: MediaPipe Holistic extracts 126 hand-landmark coordinates per frame and feeds a TensorFlow.js neural network
- Normalized landmarks by wrist position and scale for hand-size invariance; trained a binary classifier (Dropout 0.3, 50 epochs, 0.97 confidence threshold) with a custom in-browser training UI
- Segmented the user's silhouette with MediaPipe Selfie Segmentation and rendered multiple clones with smoke effects on Canvas

Leadership & Involvement

Resident Advisor, Davidson College 2026 – Present

Logistics, Hack@Davidson 2026 – Present

Event Coordinator, Davidson Entrepreneurship Club 2025 – Present

Secretary, Association for Computing Machinery (ACM), Davidson College 2026 – Present

Treasurer, Davidson International Association 2026 – Present